

Introduction to Probability

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The Math

I put a fairly lengthy pile of 'prerequisite' math on the canvas. Please tell me if you checked it out and how you found it:

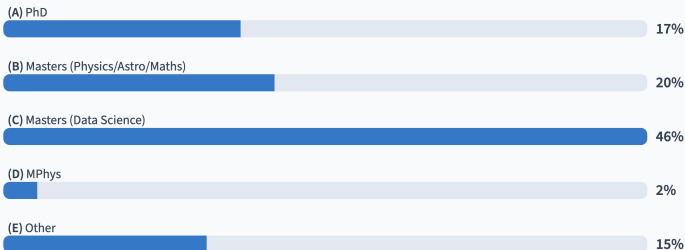
pollev.com/ilovephysics

Note: I don't expect you to be on top of the math and the python. Get comfy, and focus on whatever you have least confidence in.

For the minority of you that will find these early lectures a cakewalk, focus on your other modules (but keep your eyes open!).

Polls from Monday

DAT1: Who are you?



Polls from Monday

DAT1.3 How is your python?



Following with Python

- 1 Install marimo as per instructions on the canvas Resources & Prerequisites page.

```
pip install marimo
```

- 2 Open the canvas page for this week's material, and download the marimo notebook file Probability.py.

- 3 Run the notebook:

```
marimo edit Probability.py
```

If you didn't install marimo yet, you can view the python here:
<https://marimo.app/l/rub2g6> - best if you don't edit it today :)

Probability

- Notation
- Axioms
- Conditional Probability
- Bayes' Theorem

I include a small subset (ooh!) of terminology and symbols, for a more comprehensive list see eg [here](#).

Frequentist Probability

Example Sets:

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 2, 3, 4\}$$

The **Frequentist Probability** of A is written $P(A) = \frac{N_A}{N_S}$: the number of times A occurs in the sample divided by the number of possible outcomes in the sample. .

Probabilities:

$$P(A) = N_A/N_S = 0.5$$

$$P(B) = N_B/N_S = 0.4$$

$$P(S) = N_S/N_S = 1$$

Special Sets

$\mathbb{R}$: the set of all **Real Numbers** (all non-imaginary numbers)

$\mathbb{Q}$: the set of all **Rational Numbers** (fractions of two integers)

$\mathbb{Z}$: the set of **Integers** (negative, positive, and zero)

$\mathbb{N}$: the set of all **Natural Numbers** (positive integers and zero)

Sample Space & Subset

The **Sample Space** is denoted by **S**: the set of all possible outcomes of some experiment or operation.

A **Proper Subset** of S is written $A \subset S$: A is a subset of S but $A \neq S$.

If A is a subset of S *and* can be S, we use $A \subseteq S$.

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\}$$

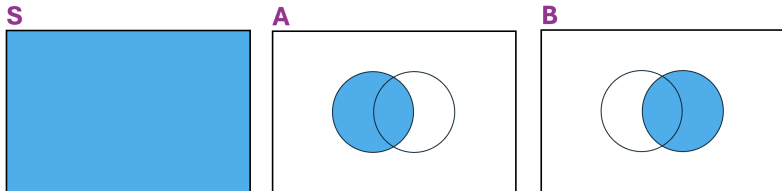
$$B = \{1, 2, 3, 4\}$$

A and B are proper subsets of S.

Venn Diagrams

Venn diagrams are helpful for visualising relationships.

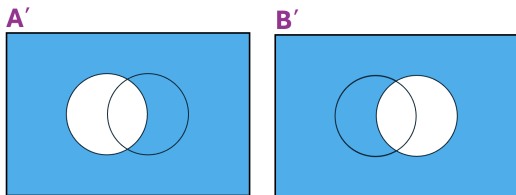
Here $A \subset S$ and $B \subset S$.



Complement

The **Complement** of a set is denoted by a prime: **A'** : this means 'not A '.

Other notations commonly used for the complement are: A^c , $\overline{A}$, and several others.



Union

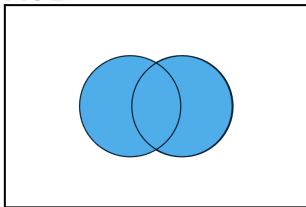
The **Union** of two sets is written: $A \cup B$: 'either A, or B, or both'.

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 2, 3, 4\}$$

$$A \cup B = \{1, 2, 3, 4, 6, 8, 10\}$$

$A \cup B$



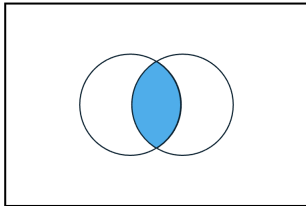
Intersection

The **Intersection** of two sets is written: $A \cap B$: 'both A and B'.

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 2, 3, 4\}$$

$$A \cap B = \{2, 4\}$$

 $A \cap B$ 

Probability of Intersection

The **Probability of Intersection** $P(A \cap B)$: the probability that both A and B are true.

This is zero when A and B are **Disjoint** (no overlap: mutually exclusive)

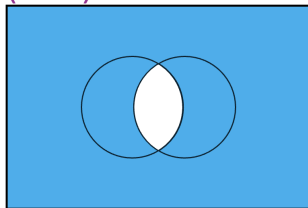
$$P(A \cap B) = 0 \text{ when } (A \cap B)'$$

Our sets:

$$A \cap B = \{2, 4\}$$

$P(A \cap B) = 0.2$: A and B are not mutually exclusive.

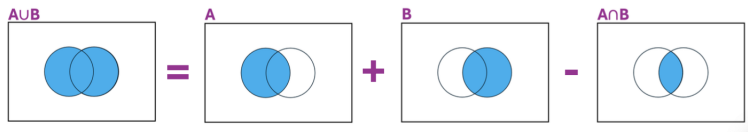
$(A \cap B)'$



Probability of Union

The **Probability of Union** $P(A \cup B)$: the probability that either A or B or both are true.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



Our sets: $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.

The Kolmogorov Axioms

1. The **Non-Negativity Axiom** states that for all events A , $P(A) \geq 0$:
'the probability of any event A must be a real number greater than zero.'
.
2. The **Normalisation Axiom** states that $P(S) = 1$: the probability of the entire sample space is one.
3. The **Countable Additivity Axiom** states that if A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$: the probability of their union is the sum of their individual probabilities.

Conditional Probability

The **Conditional Probability** $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Example:

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 2, 3, 4\}$$

$$A \cap B = \{2, 4\}$$

The number of elements in A and B is $N(A \cap B)$

The proportion of A in B is $\frac{N(A \cap B)}{N(B)}$

Relative to the original Sample Space S this is: $\frac{N(A \cap B)/N(S)}{N(B)/N(S)}$

These are probabilities: $\frac{P(A \cap B)}{P(B)}$

Given that we demand the element must exist in B , this is the probability of finding it in A .

Conditional Probability

The **Conditional Probability** $P(A|B) = \frac{P(A \cap B)}{P(B)}$.

If A and B are **Independent**, the conditional probability $P(A|B) = P(A)$.

This is because independence means that B has no effect on A and vice versa.

Our example sets are independent. $P(A|B) = P(A) = 0.5$.

Using the Conditional Probability

It is common to use conditional probabilities when we have a set of outcomes (A) and a set of choices (B).

The Conditional Probability $P(A|B) = \frac{P(A \cap B)}{P(B)}$ then gives us an answers to the question: If I choose B_i , what is the probability of the outcome A_i ?

Conditional Probability looks harmless enough, but can have some counter-intuitive results illustrated very well in the **Monty Hall Problem**, which you will look at in your homework/workshop.

The Multiplication Rule

Rearrange the Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \therefore P(A \cap B) = P(A|B)P(B)$$

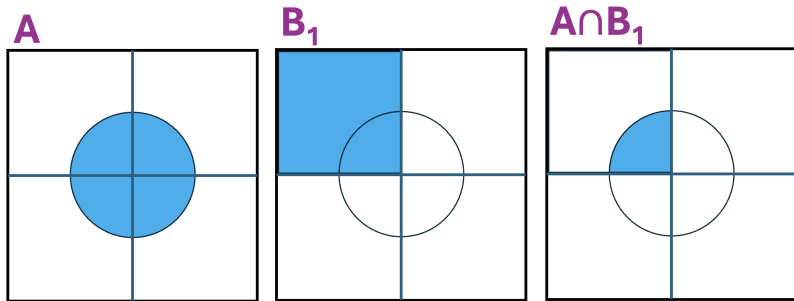
The Multiplication Rule: $P(A \cap B) = P(A)P(B)$: If A and B are independent, the probability of the intersection of A and B is equal to the product of their individual probabilities.

Example: I roll a dice twice and get two sixes:

$$P(\text{six} \cap \text{six}) = P(\text{six})P(\text{six}) = \frac{1}{6} \frac{1}{6} = \frac{1}{36}$$

The Total Probability

Referring to the picture below with $N = 4$ disjoint sets B_i intersecting with A , we can see that $P(A) = \sum_i^4 P(A \cap B_i)$.



The Total Probability

From our example: $P(A) = \sum_i^N P(A \cap B_i).$

Comparing with the Conditional: $P(A \cap B) = P(A|B)P(B)$

The **Law of Total Probability** is written:

$$P(A) = \sum_i^N P(A|B_i)P(B_i).$$

This may seem a bit contrived, but we will see soon that it is useful.

Bayes' Theorem

We have the **Conditional Probability**:

$$P(A \cap B) = P(A|B)P(B), \text{ therefore } P(B \cap A) = P(B|A)P(A)$$

Bayes' Theorem simply combines these:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

On the left we have P of A given B, and on the right P of B given A

Bayes' Theorem for our sets

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A|B) = P(A) = 0.5$$

$$P(B|A) = P(B) = 0.4$$

The so-called theorem tells us nothing! ☹

Bayes' Theorem gets interesting when we consider that it applies to **any sets A and B that satisfy Kolmogorov's Axioms.**

Bayes' Theorem for a Hypothesis

Our Hypothesis is H , for example 'My dog is a good boy'.

Our Data is X , for example 'Test results from behaviour school'.

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

$P(H|X)$: Probability of a Hypothesis, given the Data

$P(X|H)$: Likelihood of the Data, given a Hypothesis

$P(X)$: Probability of the Data

$P(H)$: Probability of the Hypothesis

The Prior Probability $P(H) = \pi(H)$

Recall, the **Frequentist Probability** of A is written $P(A) = \frac{N_A}{N_S}$: the number of times A occurs in the sample divided by the number of possible outcomes in the Sample Space.

The probability of my hypothesis $P(H) = \frac{N_H}{N_S}$: the sample space S has to be that of all Hypotheses.

This is called the **Prior Probability** because it requires some prior knowledge. In this case, the question answered by the prior would be 'What percentage of dogs are good?' This is very special so we give it a special symbol, $P(H) = \pi(H)$. Notice that it has nothing to with the test.

Bayes with Prior $\pi(H)$

Bayes is now:

$$P(H|X) = \frac{P(X|H)\pi(H)}{P(X)}$$

Using the **Law of Total probability** $P(A) = \sum_i^N P(A|B_i)P(B_i)$, we can write Bayes as:

$$P(H_i|X) = \frac{P(X|H_i)\pi(H_i)}{\sum_i^N P(X|H_i)\pi(H_i)}$$

Where H_i is my hypothesis, and the sum in the denominator is over all possible hypotheses.

Bayes' Theorem: Useful Form

$$P(H_i|X) = \frac{P(X|H_i)\pi(H_i)}{\sum_i^N P(X|H_i)\pi(H_i)}$$

$P(H_i|X)$: **Posterior Probability** of my Hypothesis being correct, given the Data.

$P(X|H_i)$: **Likelihood** of the Data, given my Hypothesis is correct

$\pi(H_i)$: **Prior Probability** of my hypothesis H_i

Denominator = $\pi(X)$: **Total probability** of the Data given any hypothesis.

Bayes' Theorem is not 'Bayesian' - it is just math

I mentioned there are two philosophies: Bayesian and Frequentist. Using Bayes' theorem does not make one a Bayesian.

A frequentist would happily use Bayes' theorem, but **not to answer the question of whether my dog is good (or any other hypothesis)**. To a frequentist, a hypothesis is either right or wrong and does not have an associated Probability Distribution.

The frequentist approach to answering this question would be to construct a **Likelihood** function $\mathcal{L}(x|\theta)$ based only on the data x , and to minimise it for a variety of different hypotheses in order to find the 'best' one.

Thanks For Your Time

Fin.